S P O O N S

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Codeline · Spark to Code 2026

Restaurant Ordering & Table Reservation

ERD & Data Mapping - Final Capstone Project team 5

Identity & Staff	Dining & Reservations	Orders & Payments	Menu & Inventory	Feedback

This document defines the full entity set, field-level schema, primary/foreign keys, and relationship mapping for the Restaurant Ordering & Table Reservation capstone, completed before any C# model or EF Core code is written - per the project's ERD-first requirement**, we will work on 10 models only**.

13 models in total: the 12 suggested entities plus MenuItemIngredient, a join table resolving the MenuItem ↔ Ingredient many-to-many relationship.

Contents

Section 2 is organized into the same five color-coded clusters used throughout this document and shown in the ERD (Section 4):

- **Identity & Staff**
- **Dining & Reservations**
- **Orders & Payments**
- **Menu & Inventory**
- **Feedback**

1. Entity Summary

The project requires a minimum of 12 models including User. This design uses 13, grouped into five functional clusters that mirror the color-coded ERD in Section 4.

IDENTITY & STAFF

User -- Customer / Staff / Admin account - basis for JWT authentication.

Staff -- Employment details for a User with Role = Staff.

DINING & RESERVATIONS

Table -- A physical dining table available for reservation.

Reservation -- A booking of a table by a user for a date/time.

ORDERS & PAYMENTS

Order -- A dine-in or takeaway order placed by a user.

OrderItem -- A line item (menu item + quantity) within an order.

Payment -- A payment (or payment attempt) against an order.

Promotion -- A discount code that can be applied to an order.

MENU & INVENTORY

MenuCategory -- Groups menu items (e.g. Starters, Mains, Drinks).

MenuItem -- A dish or drink that can be ordered.

Ingredient -- A stock item used to prepare menu items.

MenuItemIngredient -- Join table resolving the MenuItem <-> Ingredient many-to-many relationship (recipe composition).

FEEDBACK

Review — A customer rating/comment about an order and/or a menu item.

2. Entity Field Definitions

Gold rows mark the primary key. Blue rows mark foreign keys, with the referenced table.column noted in the Type / Reference column. Table headers are colored by functional cluster.

IDENTITY & STAFF

2.1 User

Key	Field	Type / Reference
PK	**Id**	int
	Name	string
	Email	string, unique
	PasswordHash	string
	Role	enum: Customer / Staff / Admin
	PhoneNumber	string
	CreatedAt	datetime

2.2 Staff

Key	Field	Type / Reference
PK	**Id**	int
FK	**UserId**	-> User.Id, unique
	Position	string
	HireDate	date
	Salary	decimal
	Shift	string

DINING & RESERVATIONS

2.3 Table

Key	Field	Type / Reference
PK	**Id**	int
	TableNumber	string
	Capacity	int
	Location	string
	IsActive	bool

2.4 Reservation

Key	Field	Type / Reference
PK	**Id**	int
FK	**UserId**	-> User.Id
FK	**TableId**	-> Table.Id
	ReservationDate	date
	ReservationTime	time
	PartySize	int
	Status	enum: Pending / Confirmed / Cancelled / Completed
	CreatedAt	datetime

ORDERS & PAYMENTS

2.5 Order

Key	Field	Type / Reference
PK	**Id**	int
FK	**UserId**	-> User.Id
FK	**TableId**	-> Table.Id (nullable)
FK	**PromotionId**	-> Promotion.Id (nullable)
	OrderType	enum: DineIn / Takeaway
	Status	enum: Pending / Preparing / Ready / Completed / Cancelled
	OrderDate	datetime
	TotalAmount	decimal

2.6 OrderItem

Key	Field	Type / Reference
PK	**Id**	int
FK	**OrderId**	-> Order.Id
FK	**MenuItemId**	-> MenuItem.Id
	Quantity	int
	UnitPrice	decimal
	Subtotal	decimal

2.7 Payment

Key	Field	Type / Reference
PK	**Id**	int
FK	**OrderId**	-> Order.Id
	Amount	decimal
	PaymentMethod	enum: Cash / Card / Online
	PaymentStatus	enum: Pending / Paid / Failed / Refunded
	PaymentDate	datetime
	TransactionRef	string

2.8 Promotion

Key	Field	Type / Reference
PK	**Id**	int
	Code	string, unique
	Description	string
	DiscountPercent	decimal
	StartDate	date
	EndDate	date
	IsActive	bool

MENU & INVENTORY

2.9 MenuCategory

Key	Field	Type / Reference
PK	**Id**	int
	Name	string
	Description	string
	DisplayOrder	int

2.10 MenuItem

Key	Field	Type / Reference
PK	**Id**	int
	Name	string
	Description	string
	Price	decimal

Key	Field	Type / Reference
	ImageUrl	string
	IsAvailable	bool
FK	**MenuCategoryId**	-> MenuCategory.Id

2.11 Ingredient

Key	Field	Type / Reference
PK	**Id**	int
	Name	string
	UnitOfMeasure	string
	QuantityInStock	decimal
	ReorderLevel	decimal

2.12 MenuItemIngredient

Key	Field	Type / Reference
PK	**Id**	int
FK	**MenuItemId**	-> MenuItem.Id
FK	**IngredientId**	-> Ingredient.Id
	QuantityRequired	decimal

FEEDBACK

2.13 Review

Key	Field	Type / Reference
PK	**Id**	int
FK	**UserId**	-> User.Id
FK	**OrderId**	-> Order.Id (nullable)
FK	**MenuItemId**	-> MenuItem.Id (nullable)
	Rating	int (1-5)
	Comment	string
	CreatedAt	datetime

3. Relationship Mapping

One-to-many relationships are enforced via a foreign key on the "many" side referencing the primary key of the "one" side. The single many-to-many relationship (MenuItem ↔ Ingredient) is resolved with a join table carrying both foreign keys.

From	To	Cardinality	Description	Mapping
User	**Reservation**	**1 : N**	A user can make many reservations.	User.Id -> Reservation.UserId
User	**Order**	**1 : N**	A user can place many orders.	User.Id -> Order.UserId
User	**Review**	**1 : N**	A user can write many reviews.	User.Id -> Review.UserId
User	**Staff**	**1 : 1**	A staff-role user has one employment record.	User.Id -> Staff.UserId (unique)
Table	**Reservation**	**1 : N**	A table can be booked in many reservations (over time).	Table.Id -> Reservation.TableId
Table	**Order**	**1 : N (optional)**	A dine-in order is assigned to one table; takeaway leaves this null.	Table.Id -> Order.TableId
MenuCategory	**MenuItem**	**1 : N**	A category groups many menu items.	MenuCategory.Id -> MenuItem.MenuCategoryId
MenuItem	**OrderItem**	**1 : N**	A menu item can appear as a line item on many orders.	MenuItem.Id -> OrderItem.MenuItemId
Order	**OrderItem**	**1 : N**	An order is made up of many order line items.	Order.Id -> OrderItem.OrderId
Order	**Payment**	**1 : N**	An order can have multiple payment attempts/records.	Order.Id -> Payment.OrderId
Promotion	**Order**	**1 : N (optional)**	A promotion code can be applied to many orders.	Promotion.Id -> Order.PromotionId
Order	**Review**	**1 : N (optional)**	A completed order can receive reviews.	Order.Id -> Review.OrderId
MenuItem	**Review**	**1 : N (optional)**	A menu item can receive reviews directly.	MenuItem.Id -> Review.MenuItemId
MenuItem	**Ingredient**	**N : N**	A menu item uses many ingredients; an ingredient serves many menu items.	Via MenuItemIngredient (MenuItemId, IngredientId)

3.1 Notes on optional (nullable) foreign keys

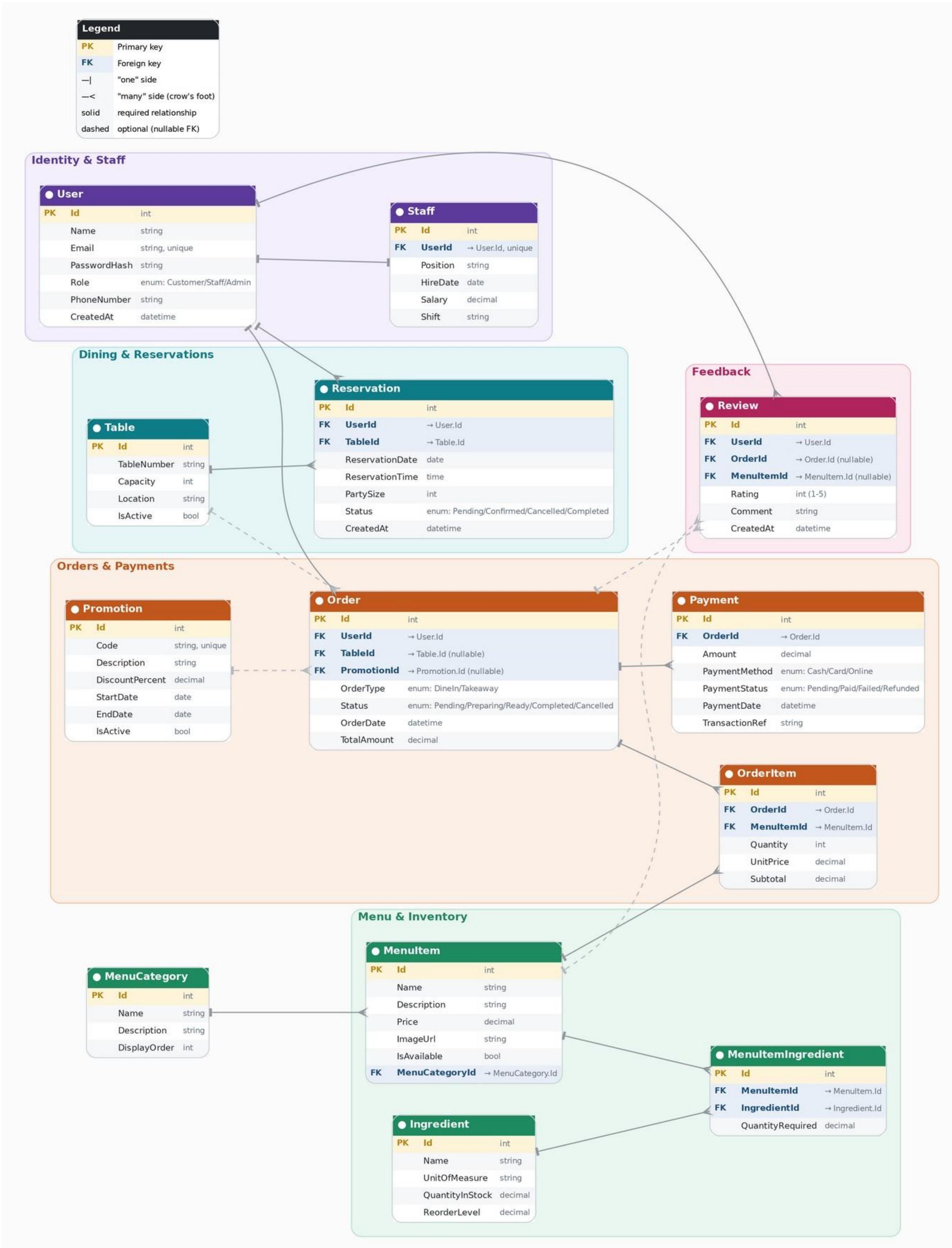
Order.TableId is nullable because takeaway orders are not assigned to a table. Order.PromotionId is nullable because a discount code is not required. Review.OrderId and Review.MenuItemId are both nullable - a review may reference an order, a menu item, or both, but always references a User.

3.2 Legend

Symbol	Meaning
PK	Primary key of the table
FK	Foreign key referencing another table's PK
— |	The "one" side of a relationship
—<	The "many" side of a relationship (crow's foot)
solid line	Required relationship (non-nullable FK)
dashed line	Optional relationship (nullable FK)

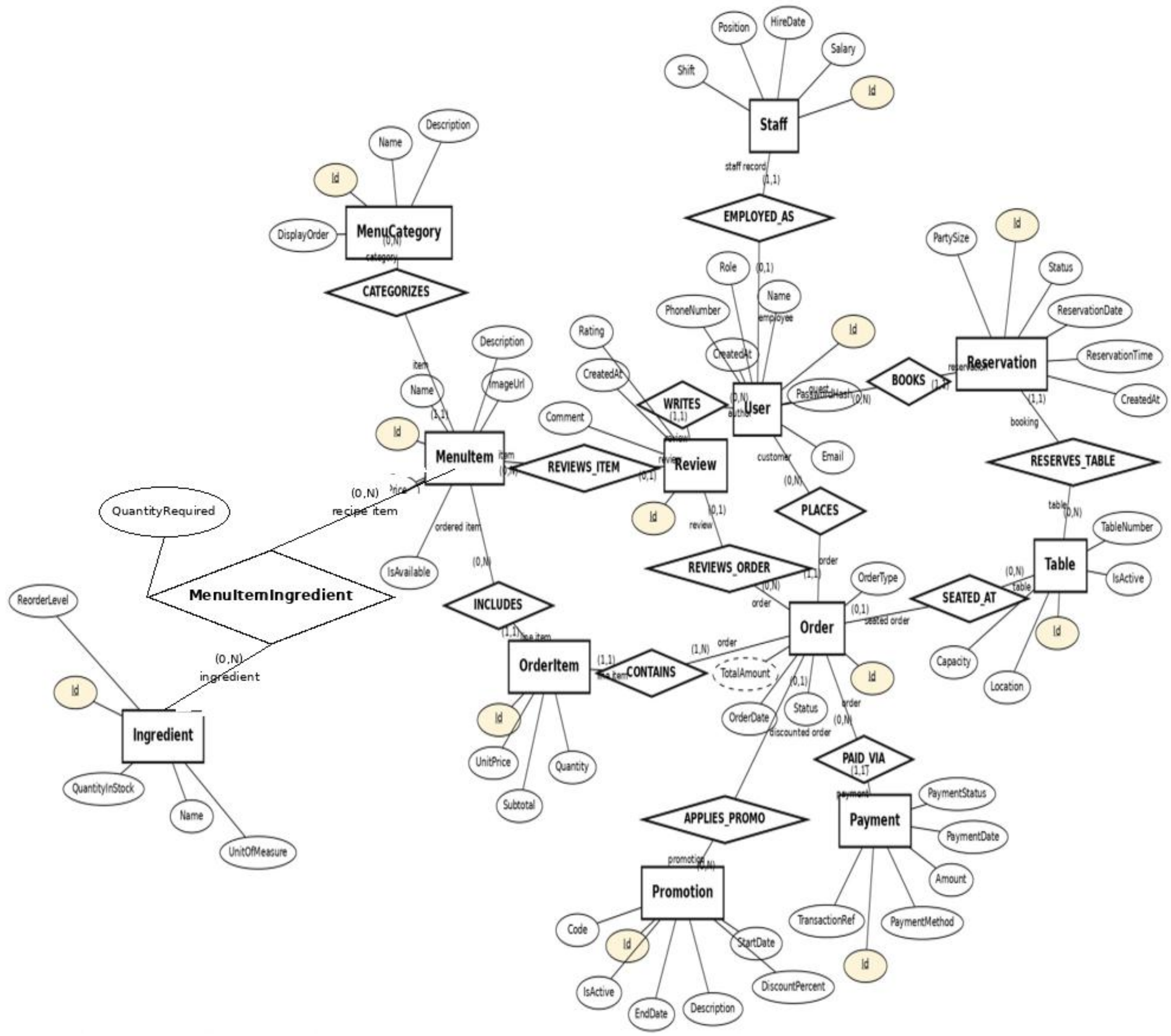
4. ERD Diagram (crow's-foot / cluster notation)

Entities are grouped into color-coded functional clusters. Solid lines are required foreign keys; dashed lines are optional (nullable) foreign keys. Line ends follow crow's-foot notation -- a bar for "one," a crow's foot for "many."



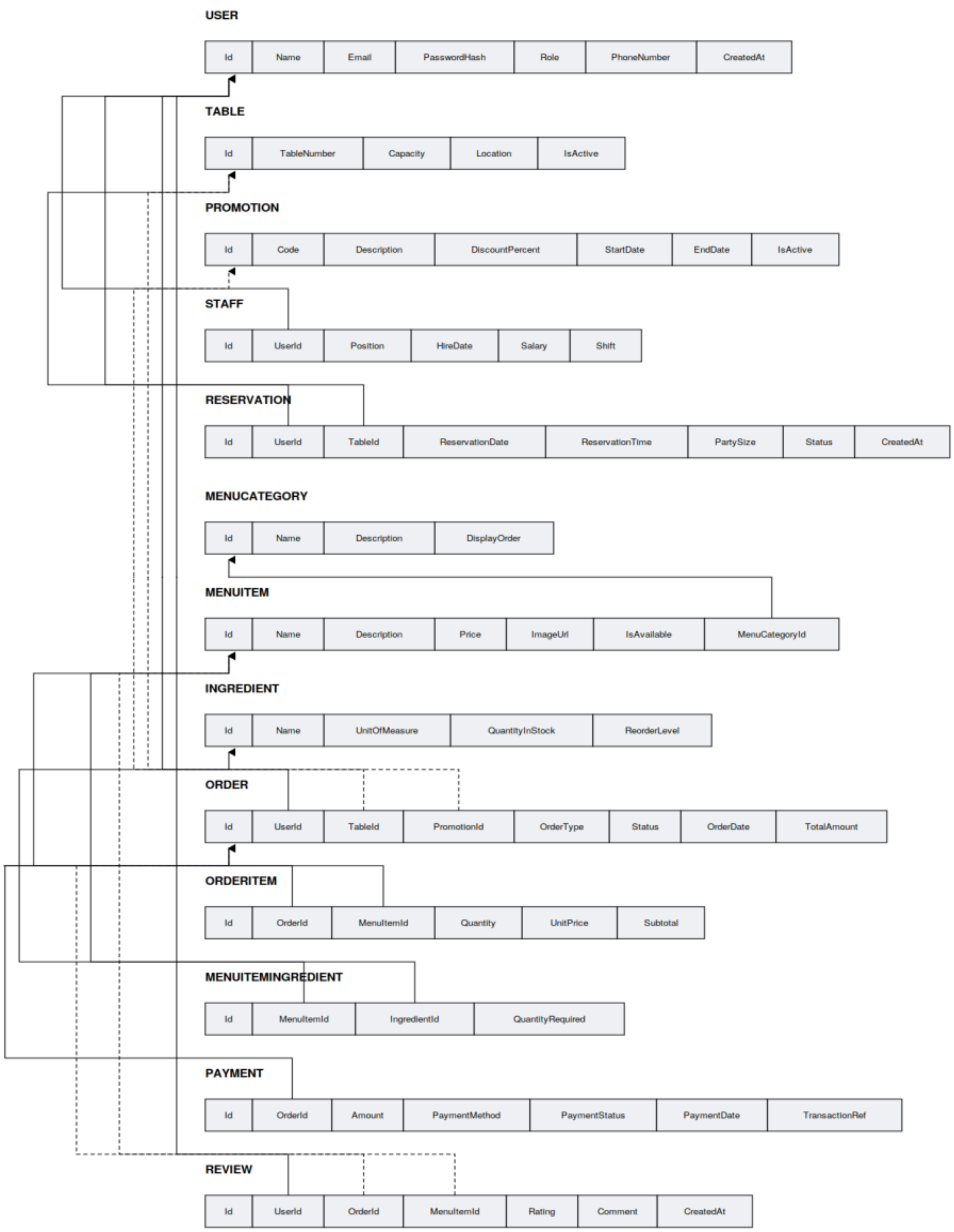
4b. ERD Diagram — Classic (Chen) Notation

Same schema, drawn in the traditional Chen ER style: rectangles are entities, ellipses are attributes (underlined = primary key, dashed = derived), and diamonds are relationships, each edge carrying a role name and a (min,max) structural participation constraint. MenuItemIngredient is modeled as a weak/associative entity (double rectangle), resolving the MenuItem ↔ Ingredient many-to-many relationship through two identifying relationships (double diamonds): REQUIRES and USED_IN.



5. Mapping Diagram

The ERD-to-relational mapping view lists every table's columns in a single flow, with arrows pointing from each foreign key to the primary key it references. Underlined columns are primary keys; dashed arrows mark optional (nullable) foreign keys.



The Mapping Diagram after deleting Staff and Promotion